Prompt-Variant Diagnostic: Short Analysis

Does tokenization fragmentation predict miscalibration in Bangla LLMs? Held-out dev-pool results for three models

Bottom line. One uniform setting, **few-shot (4 examples), chat template off, no debiasing**, passes both validity gates for all three models, so it can be the primary inference setting without per-model prompt differences. Two more uniform settings also pass for all three: few-shot with cyclic4 debiasing and answer_cue with cyclic4. The collapse seen in the full run (answer_cue, no debiasing) is reproduced here. Gate passes only show the output is usable; they do not show the small models can do the task.

1. Setup

- Each model was probed under **12 inference settings**: prompt variant (plain, answer_cue, few_shot with 4 balanced examples) × chat template (off/on) × debiasing (none, or cyclic4 = average over four cyclic option rotations).
- **60 held-out dev items per language** (disjoint from the evaluation items and the few-shot examples). Bangla = BnMMLU, English = subject-matched MMLU.
- **Validity gates** (both languages): mean option mass ≥ 0.50 and top-letter share ≤ 0.80. Settings are judged on the gates only; accuracy is reported but never used for selection.
- *Model identity is inferred from file order* (Llama, TigerLLM, TituLLM) and is consistent with prompt lengths: Llama and TituLLM share one tokenizer (about 233 Bangla prompt tokens), TigerLLM (Gemma-based) needs about 76.

2. Which settings pass the gates

Setting (variant / chat / debias)	Llama-3.2-3B	TigerLLM-1B	TituLLM-1B
plain / chat off / none	FAIL (mass, collapse)	FAIL (collapse)	FAIL (mass, collapse)
plain / chat off / cyclic4	FAIL (mass)	PASS	FAIL (mass)
plain / chat on / none	PASS	FAIL (mass, collapse)	FAIL (mass, collapse)
plain / chat on / cyclic4	PASS	FAIL (mass)	FAIL (mass)
answer_cue / chat off / none	PASS	FAIL (collapse)	FAIL (collapse)
answer_cue / chat off / cyclic4	PASS	PASS	PASS
answer_cue / chat on / none	PASS	FAIL (mass, collapse)	FAIL (collapse)
answer_cue / chat on / cyclic4	PASS	FAIL (mass)	PASS
few_shot / chat off / none	PASS	PASS	PASS
few_shot / chat off / cyclic4	PASS	PASS	PASS
few_shot / chat on / none	FAIL (mass)	FAIL (mass, collapse)	PASS
few_shot / chat on / cyclic4	FAIL (mass)	FAIL (mass)	PASS

Table 1. Gate outcome per setting. Passes: Llama 8/12, TigerLLM 4/12, TituLLM 6/12. Yellow setting labels pass for all three models. "mass" = mean option mass < 0.50 in at least one language; "collapse" = one letter > 80% of predictions.

3. Findings

- **The earlier collapse is reproduced.** With answer_cue and no debiasing, TituLLM predicts one letter for 93% of Bangla items and TigerLLM for 87% of English items (full run: 92.2% and 85.3%). Zero-shot prompts leave small models with a strong letter prior.
- **Few-shot fixes the output format.** Option mass rises to ≥ 0.97 for every model in both languages. Raw letter bias is reduced but not removed (Table 2, Figure 1); cyclic4 pulls it down to about 0.28–0.33.
- **The chat template is model-specific.** It helps Llama (plain prompt: option mass 0.48 → 0.90) but breaks TigerLLM: every chat-on row has option mass 0.01–0.18, so the instruction-tuned Gemma model does not start its reply with an option letter. For TituLLM the chat-on rows are nearly identical to chat-off (prompt length differs by one token), so its template appears to be a no-op.
- **Fragmentation shows up in the prompts.** Bangla few-shot prompts are about 1,225 tokens for Llama/TituLLM versus 370 for TigerLLM; English prompts are about 505–509 for all. Llama and TituLLM share an identical tokenizer, so that pair cannot contrast low versus high fragmentation.

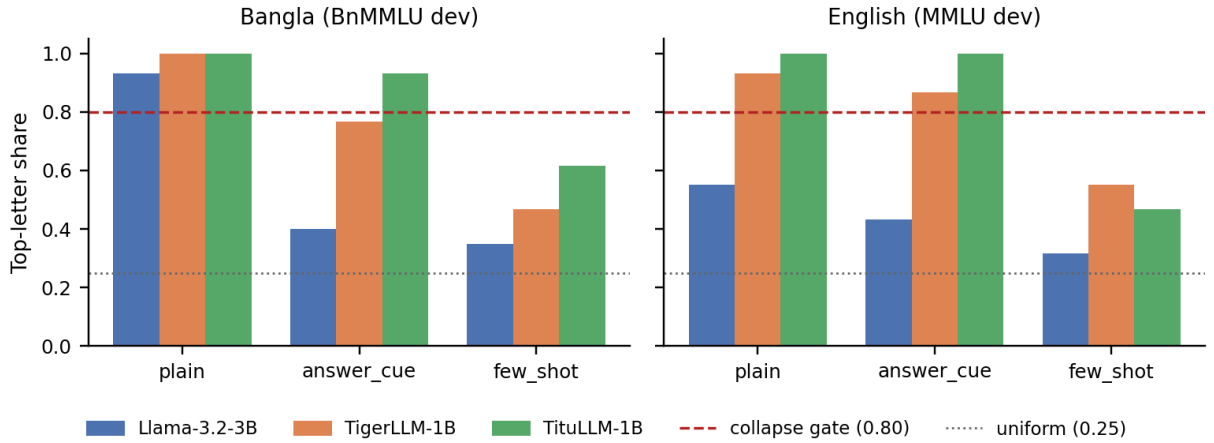


Figure 1. Raw top-letter share (chat off, no debiasing). Dashed line: collapse gate (0.80); dotted line: uniform expectation (0.25). Few-shot is the only variant with all six bars below the gate.

Model	Option mass BN / EN	Top-letter share BN / EN	Dev accuracy (%) BN / EN	BN prompt tokens
Llama-3.2-3B	0.973 / 1.000	0.35 / 0.32	50.0 / 63.3	1225
TigerLLM-1B	0.991 / 0.972	0.47 / 0.55	33.3 / 35.0	370
TituLLM-1B	0.990 / 0.999	0.62 / 0.47	33.3 / 21.7	1225

Table 2. Recommended uniform setting (few_shot, chat off, no debiasing), 60 dev items per language. Accuracy is informational only (standard error about 5.6 points at n = 60; chance = 25%).

4. Caveats

- **Small sample.** With 60 items per language, gate decisions near a threshold are unstable (for example Llama plain, chat off: mass 0.479 in Bangla). The 1,500-item full run decides final validity.
- **Passing the gates is not the same as being informative.** Dev accuracy for TituLLM in English (21.7%) is at chance level (standard error about 5.6 points) and TigerLLM is about 33–35%. Calibration analyses are only meaningful for runs whose full-run accuracy is clearly above 25%.
- **Debiasing changes the meaning of confidence.** cyclic4 reports the average over four option rotations, so it should be a labelled sensitivity run with raw and debiased calibration reported side by side.
- **Per-model rules would introduce a confound.** The notebook's own preference order would pick plain + chat on for Llama but few-shot for the others; a uniform setting avoids prompt differences between models.

5. Recommended next steps

- Archive the old Llama run (answer_cue + chat template) and re-run **all three models** with few_shot (k = 4), chat off, no debiasing as the primary setting.
- Run few_shot + cyclic4 for all three as a sensitivity analysis; report raw and debiased ECE and Brier.
- After the full run, check each run's validity flag and accuracy against chance before any fragmentation analysis; document per-model settings and the tokenizer families (Llama/TituLLM identical, TigerLLM Gemma-based) in the methodology.